Business and System Requirements Document

Beta cinema booking reliability enhancement initiative

05/2026

Document revisions

Date	Version Number	Document Changes
22/05/2026	1.0	Initial document

PART I - BUSINESS ANALYSIS

1 Introduction

1.1 Project summary

1.1.1 Objectives

The project is conducted with the aim of optimizing the ticket booking system within the Beta Cinema application. By re-engineering the seat-locking mechanism and asynchronous payment handling, the project targets to eliminate transactional data loss, lower customer churn rates, and maximize seat occupancy efficiency for theater operations

1.1.2 Problem statements

Problem 1 (self-locking seat defect): Users are blocked from re-selecting their own previously chosen seats if they navigate back or exit during the checkout phase. The system retains a "Blind Lock" on the seats for 10 minutes without validating whether the returning user is the original holder.

Problem 2 (unissued tickets post-payment): Users complete successful monetary deductions via third-party payment gateways (e.g., Momo, Zalo pay), but zero ticket issuance.

1.1.3 Goal of project

To architect a resilient Booking State Machine and user-bound data structure that guarantees transactional integrity-ensuring that:

- One-and-only-one customer can claim a seat successfully.
- Authentic users can seamlessly resume booking their held seats upon back-navigation.
- Zero data discrepancy occurs between payment authorization and ticket fulfillment.

1.2 Project scope

1.2.1 In scope functionality

- Bind seat with user/session: Store the Seat_ID together with the User_ID or Session_ID while the seat is being held for 10 minutes.
- Keep user session during payment flow: Ensure the user session remains the same when moving between the Beta Cinema app and the external payment gateway.
- Add booking processing status: Create a temporary booking status while ticket issuance is in progress after payment confirmation.
- Add release seat API: When users cancel payment, go back, or leave the booking flow, the system should immediately release the held seats instead of waiting 10 minutes.
- Improve payment callback handling: Add retry and recovery mechanisms to ensure successful payment confirmations are received and processed correctly.

- Handle late payment confirmation: Prevent automatic ticket issuance when successful payment confirmation is received after the 10-minute holding period and route the transaction for reconciliation or refund processing.

1.2.2 Out of scope functionality

- UI/UX redesign: Changing the visual design or layout of the booking and payment screens is not included in this project.
- Payment gateway performance improvement: Improving the speed or technical performance of third-party payment providers such as MoMo or VNPAY is not included.
- Manual refund/support tools: Building admin tools for customer support teams to manually handle failed bookings or refunds is not included.

1.3 Constrait

1.3.1 Timeline constraint

The system must be fully deployed before the peak movie season to minimize operational risks during high-traffic periods.

1.3.2 Compliance constraint

The system must securely protect user payment information and comply with Vietnam's Personal Data Protection regulations (Decree No. 13/2023/ND-CP).

1.3.3 Resource constraint

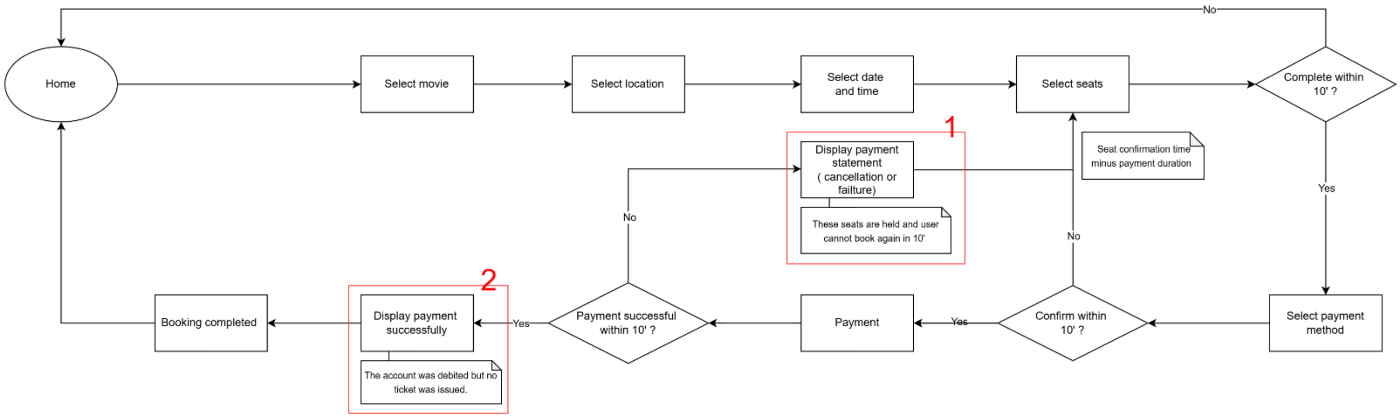
System development, testing, and deployment must be completed by the internal technical team within the approved infrastructure and project budget.

1.4 Definitions

STT	Term	Definition
1	Delayed ticket issuance	Ticket generation is not completed within 3 seconds after successful payment confirmation
2	Holding period	The period during which selected seats are reserved for a customer
3	Booking session	A temporary session associated with reserved seats and booking information

2 Current process overview

2.1 Current user flow (As-Is)



2.2 Analysis problem

Based on the current booking flow, two major issues have been identified in the system. These issues mainly relate to seat reservation handling and ticket issuance after payment completion.

2.2.1 Problem 1 (self-locking seat defect):

This issue is identified at the area marked as (1) in the current user flow.

Description: When a user enters the payment screen but cancels, navigates back, or disconnects, the system continues to lock their selected seats. If the same user tries to re-select those seats immediately, the system blocks them for the entire 10-minute holding window.

Possible root cause 1: No immediate unlock trigger

- Scenario: The user cancels payment, but the seat remains locked.

- Root Cause: Although the system detects the interrupted payment transaction, it fails to release the associated seat lock immediately. As a result, the seats remain locked until the 10-minute holding timer expires, preventing even the same user from selecting them again during that period.

Possible root cause 2: Anonymous seat locking

- Scenario: The same user goes back or disconnects but cannot select their own seats.

- Root Cause: The system only records that the seat is locked, but fails to record who locked it. Because it does not recognize the returning user as the original owner, it blocks them.

2.2.2 Problem 2 (unissued tickets post-payment):

This issue is identified at the area marked as (2) in the current user flow.

Description: The user is successfully debited by the third-party payment gateway, but the system fails to internally process the transaction or issue a digital ticket, abruptly redirecting the user back to the Home screen without any confirmation or error notice.

Possible root cause 1: Lost payment signal

- Scenario: Money is deducted, but the user is kicked to the Home screen without a ticket.

- Root Cause: The connection between the payment gateway (MoMo/VNPAY) and Beta's server drops due to high traffic or network errors. The gateway takes the money but Beta's server misses the success confirmation signal.

Possible root cause 2: Payment timeout mishandling

- Scenario: The user completes the payment at the last seconds, but no ticket is issued.

- Root Cause: If bank processing takes too long and the confirmation arrives late (e.g., at minute 10:10), Beta's system has already automatically cancelled the order at minute 10:00 and rejects the late payment confirmation.

Possible root cause 3: Database saving failure

- Scenario: Payment is successful, but the system still fails to generate the ticket.

- Root Cause: High system traffic causes an internal database error (Error 500) when trying to save the ticket and update the seat status at the same time. The process fails halfway: money is taken, but the ticket cannot be saved.

3 Solution

3.1 Proposed solutions

3.1.1 Proposed solutions for problem 1 (self-locking seat defect)

Solution 1 — Immediate seat release on explicit cancellation

The system should automatically release reserved seats immediately when the user explicitly cancels the payment process or receives a failed transaction result from the gateway. This ensures instant seat availability for other customers and prevents unnecessary locking.

Solution 2 — Session-bound temporary seat ownership

For unexpected interruptions (e.g., network disconnection, navigating back, or app closure), the system should temporarily bind the reserved seats to the original user's session ID or booking token. This allows the same user to re-access and continue booking their selected seats within the 10-minute holding window, while preventing seats from remaining locked without active user engagement.

3.1.2 Proposed solutions for problem 2 (unissued tickets post-payment)

Solution 1 — Payment callback retry mechanism

The system should implement a retry mechanism for failed payment callback processing if the initial payment gateway callback is not successfully received. This minimizes the risk of successful transactions being lost due to transient network drops.

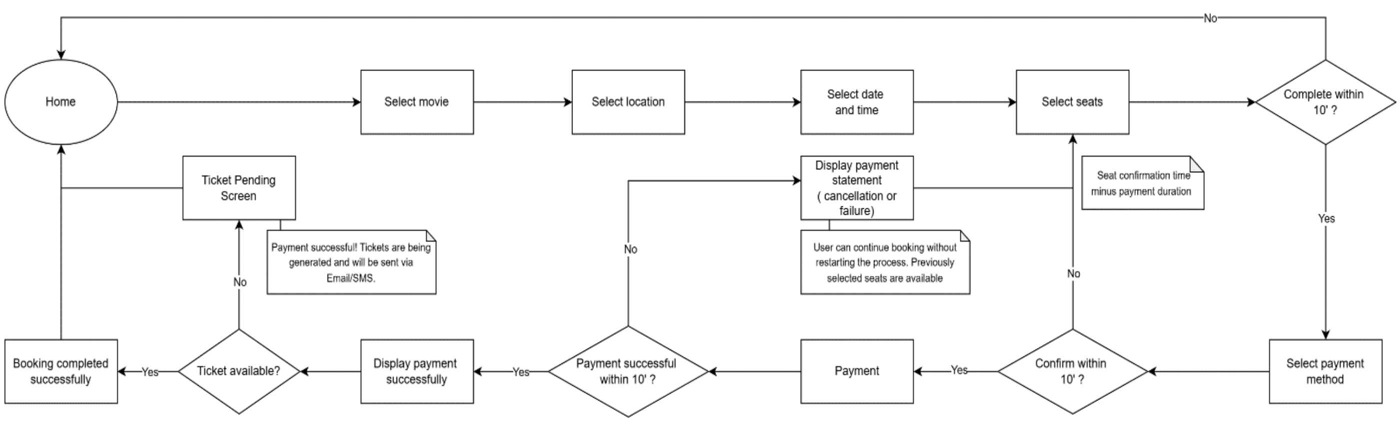
Solution 2 — Asynchronous ticket issuance with pending state UI

The system should process ticket issuance asynchronously after payment confirmation and temporarily display a processing status to the user. If database timeouts or processing failures occur, the booking request should be placed into a recovery queue while the user receives a notification confirming that ticket issuance is still in progress via Email or SMS.

Solution 3 — Error logging, alerting, and automated recovery

The system should comprehensively record payment and booking anomalies in error logs, trigger real-time alerts for the technical team, and initiate automated recovery workflows (such as background reconciliation). This ensures robust transaction traceability and rapid issue resolution.

3.2 To-be user flow



4 Business rules

4.1 Seat reservation rules

4.1.1 Business rule 1.1

ID: BR-01

TITLE: Maximum seat holding time

DESC: Selected seats shall be held for a maximum of 10 minutes per transaction. After 10 minutes, the seats are automatically released for others.

4.1.2 Business rule 1.2

ID: BR-02

TITLE: Session-bound seat ownership

DESC: During the active holding period, reserved seats shall remain associated with the original booking session.

4.1.3 Business rule 1.3

ID: BR-03

TITLE: Immediate seat release

DESC: Reserved seats shall be immediately released if the user cancels the transaction.

4.1.4 Business rule 1.4

ID: BR-04

TITLE: Preserve seats on payment failure

DESC: The system shall retain reserved seats within the active holding period when a payment attempt fails, allowing the customer to retry payment.

4.2 Payment & ticketing rules

4.2.1 Business rule 2.1

ID: BR-05

TITLE: Ticket issuance condition

DESC: Digital tickets shall only be generated and delivered after successful payment confirmation.

4.2.2 Business rule 2.2

ID: BR-06

TITLE: Late payment handling

DESC: If a successful payment confirmation arrives after the 10-minute window has expired, the system must not issue a ticket automatically. The transaction shall be flagged for reconciliation, manual review, or refund processing.

4.2.3 Business rule 2.3

ID: BR-07

TITLE: Technical error transparency

DESC: If payment is successful but ticket issuance fails due to a system error, the user shall be shown a pending status notification instead of being redirected without confirmation.

4.2.4 Business rule 2.4

ID: BR-08

TITLE: Backup notification channel

DESC: For any delayed or pending transactions, the final status (success or failure) shall be sent to the user via Email or SMS as soon as the issue is resolved.

PART II - BUSINESS REQUIREMENTS

5 Functional requirements

5.1 Functional requirements for problem 1 (self-locking seat defect)

5.1.1 Functional requirement 1.1

ID: FR-01

TITLE: Immediate seat release on payment cancellation

DESC: The system shall immediately release reserved seats when the user cancels the payment process.

RAT: To prevent seats from remaining unnecessarily locked during the holding period.

DEP: None

MAPPING: BR-03

PRIORITY: High

5.1.2 Functional requirement 1.2

ID: FR-02

TITLE: Preserve the seats on payment failure

DESC: The system shall preserve reserved seats when the payment gateway returns a failed transaction result.

RAT: Allow users to continue booking the selected seats without restarting the process.

DEP: None

MAPPING: BR-04

PRIORITY: High

5.1.3 Functional requirement 1.3

ID: FR-03

TITLE: Preserve and restore interrupted booking session

DESC: The system shall allow the original user to resume an interrupted booking session using the existing session ID or booking token. If the 10-minute holding window expires without the session being restored, the system shall automatically release the reserved seats for other users.

RAT: Allow users to continue booking without restarting the process and prevent seats from being unnecessarily locked.

DEP: None

MAPPING: BR-01, BR-02

PRIORITY: High

5.2 Functional requirements for problem 2 (unissued tickets post-payment)

5.2.1 Functional requirement 2.1

ID: FR-04

TITLE: Validate payment confirmation before booking completion

DESC: The system shall verify successful payment confirmation before completing the booking process. In case the initial payment gateway callback is not received due to network drops, the system shall implement a retry mechanism to re-process the payment status.

RAT: Reduce the risk of payment success without ticket issuance.

DEP: None

MAPPING: BR-05

PRIORITY: High

5.2.2 Functional requirement 2.2

ID: FR-05

TITLE: Display booking processing status

DESC: The system shall display a booking processing status after successful payment confirmation when ticket issuance is delayed or temporarily pending.

RAT: Inform users about ticket issuance.

DEP: FR-04

MAPPING: BR-07

PRIORITY: High

5.2.3 Functional requirement 2.4

ID: FR-06

TITLE: Send booking confirmation and ticket information

DESC: For delayed or pending transactions, the system shall send booking confirmation via Email or SMS after successful ticket generation.

RAT: Ensure users receive official booking confirmation and ticket details.

DEP: FR-04, FR-05

MAPPING: BR-08

PRIORITY: High

5.2.4 Functional requirement 2.5

ID: FR-07

TITLE: Payment anomaly logging and real-time alerting

DESC: The system shall comprehensively record all booking and payment anomalies (e.g., timeout after minute 10:00, Database Error 500) into error logs and trigger immediate real-time alerts to the technical support team.

RAT: Improve system reliability and reduce unresolved transaction failures.

DEP: None

MAPPING: BR-06

PRIORITY: Medium

6 Non-functional requirements

6.1 Reliability & availability

6.1.1 Non-funtional requirement 1.1

ID: NFR-01

TITLE: Payment callback retry policy

DESC: The system shall retry failed payment synchronization attempts at least 5 times before marking the transaction for manual recovery.

6.1.2 Non-funtional requirement 1.2

ID: NFR-02

TITLE: Transactional data consistency

DESC: The system shall ensure transactional consistency so that seat reservation updates and ticket generation are completed successfully together or rolled back together in case of failure.

6.2 Performance & scalability

6.2.1 Non-funtional requirement 2.1

ID: NFR-03

TITLE: Async ticket issuance latency

DESC: The system shall complete ticket generation within 3 seconds under peak traffic conditions.

6.2.2 Non-funtional requirement 2.2

ID: NFR-04

TITLE: Pending ticket recovery time

DESC: The system shall complete ticket generation for pending bookings within 15 minutes after successful payment confirmation.

6.2.3 Non-functional requirement 2.3

ID: NFR-05

TITLE: Seat release response time

DESC: The system must process and complete the explicit seat release API call within less than 1 second after the user clicks "Cancel".

6.3 Security

6.3.1 Non-functional requirement 3.1

ID: NFR-06

TITLE: Booking token security & expiration

DESC: Session IDs and booking tokens shall expire automatically after the holding period ends.

6.4 Observability & alerting

6.4.1 Non-functional requirement 4.1

ID: NFR-07

TITLE: Real-time alerting latency

DESC: The system shall trigger real-time alerts for critical payment or internal server failures within 60 seconds of occurrence.

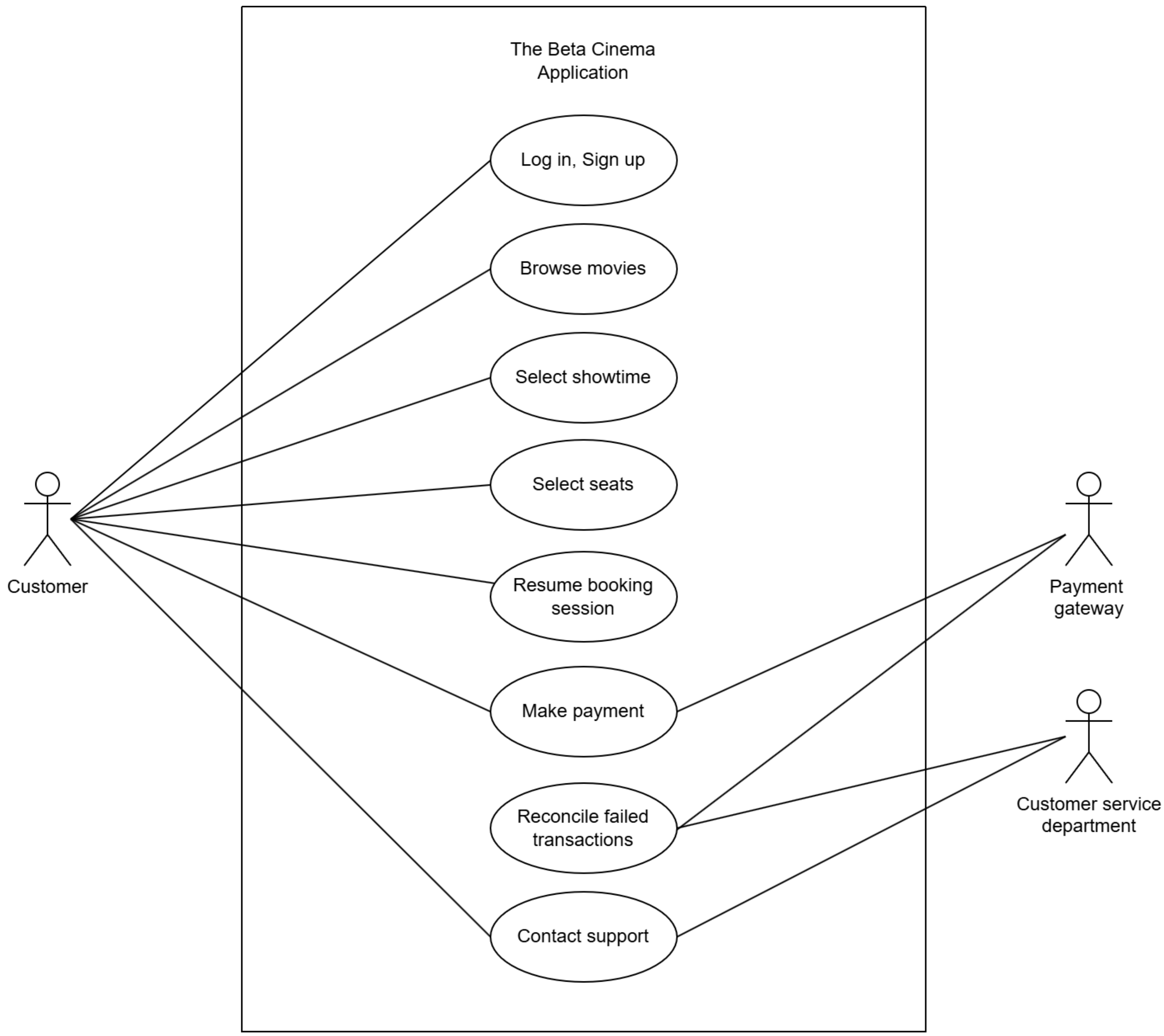
7 Use case specification

Use case ID	UC-01
Use case name	Movie ticket booking
Description	Allows customers to select movie tickets, complete payment, resume interrupted booking sessions, and receive ticket confirmation
Actors	Customer, payment gateway
Priority	Must have
Trigger	Customer wants to book movie tickets through the application
Pre-conditions	• The customer has a valid account • Internet connection is available • Selected movie schedules and seats are available for booking
Post-conditions	• The booking is successfully completed • Digital ticket information is generated • The booking transaction is recorded successfully in the system

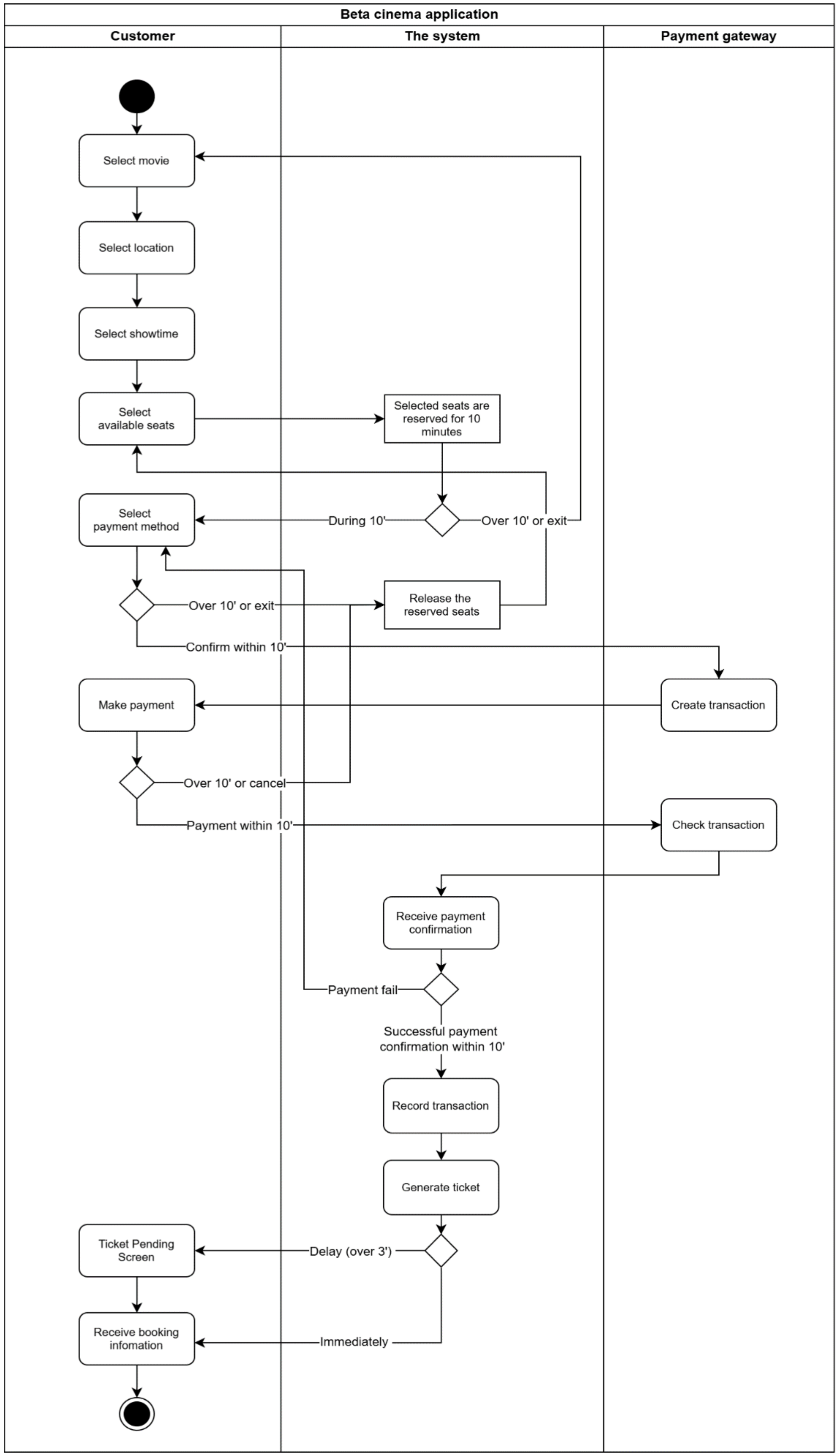
Basic flow	1. Customer selects a movie 2. Customer selects cinema location, date, and time 3. Customer selects available seats 4. The system temporarily reserves selected seats within the holding period 5. Customer selects a payment method and confirms the booking 6. Customer completes payment successfully within the allowed time 7. The payment gateway returns successful payment confirmation 8. The system validates payment information 9. The system initiates ticket generation processing 10. The system displays booking confirmation
Alternative Flow	5a. Customer backs to previous page 5a1. The system imediately releases the reserved seats 5a2. Use Case resumes at Step 3 6a. Customer exits the application, disconnects during booking 6a1. The system preserves the booking session within the holding period 6a2. Customer reconnects and Use Case resumes at Step 6 6b. The customer attempts to make a payment, but the transaction fails 6b1. The system retains the booking session and reserved seats within the active holding period 6b2. Use Case resumes at Step 6 6c. Customer cancels payment 6c1. The system immediately releases the reserved seats 6c2. Use Case resumes at Step 3 9a. Ticket generation is temporarily delayed after successful payment 9a1. The system displays a ticket processing status screen 9a2. The system continues processing ticket generation asynchronously 9a3. The system sends ticket information via Email or SMS after processing is completed

Exception Flow	5b. Booking session timeout (customer abandons or fails to reconnect) 5b1. The 10-minute holding period expires without payment completion or session resumption 5b2. The system automatically releases the reserved seats 5b3. Use case ends. 6d. Payment confirmation is not received within the holding period 6d1. The system cancels the booking transaction and releases reserved seats 6d2. If payment confirmation is received after the holding period expires, the system routes the transaction to the Reconciliation Queue for review or refund. 6d3. Use case ends 6e. Successful payment confirmation arrives after the 10-minute window has expired 6e1. The system verifies that the booking session has expired. 6e2. The system does not issue tickets automatically and flags the transaction for reconciliation. 6e3. The system records the anomaly and triggers a real-time alert to the technical support team. 6e4. The system notifies the customer that the transactionis is being reviewed. 6e5. Use case ends.

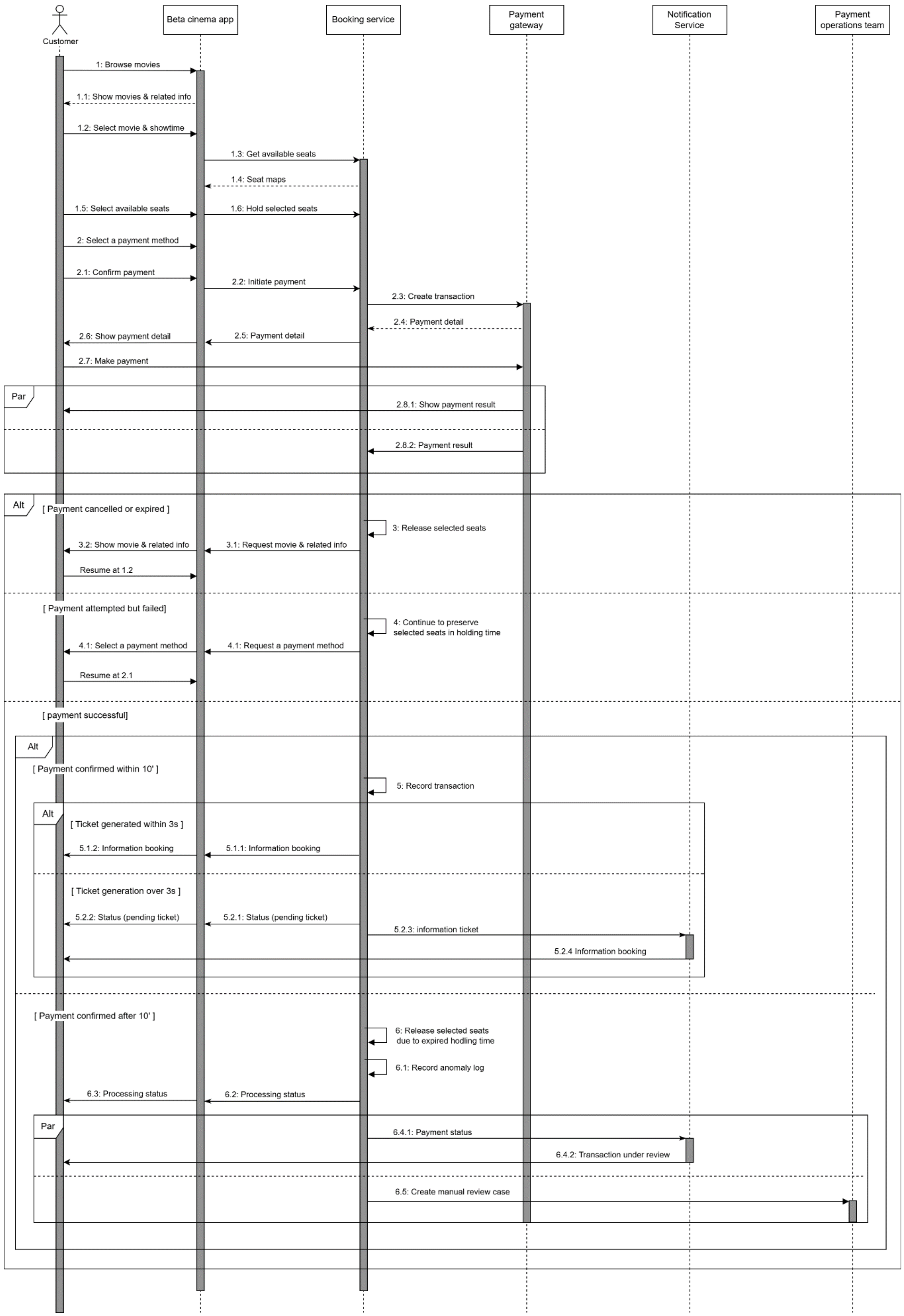
8 Use Case Diagram



9 Activity diagram



10 Sequence diagram



11 Entity relationship diagram